

Soren Antebi

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EXPERIENCE

Fraunhofer Institute (IAIS)

Bonn, DE

Machine Learning Engineer

June 2024 – Present

- Developed deep learning solutions with industry clients (Daimler, Lufthansa, MAN) for automation and quality control optimization, reducing manual inspection time by over 50%+ (OpenCV, PyTorch, C++).
- Designed OpenEvolve-based research Agent for evolutionary autoresearch on vision tasks (GPT-5.4, Minimax 2.5).
- Led 3+ machine vision projects end-to-end, driving system/hardware design, model selection and development, and client collaboration across CNN, ViT, and multimodal solutions, with CI/CD and Git.
- Engineered a C++/Qt system for reflective surface analysis, involving video-to-image reconstruction and object detection using custom deep learning models; configured via MQTT across 15+ cameras and 6 edge devices.
- Deployed ONNX models for C++ inference with OpenVINO and ONNXRuntime, reducing latency by 20%.
- Collaborated in a team of 4 to secure €100,000 in funding at an institute-wide hackathon for a computer vision robotics project.

Imperial College London

London, UK

MSc Research Assistant in Machine Learning, Supervisor: Dr. Ben Glocker

March 2023 – October 2023

- Trained and analyzed 50+ deep learning models for disease detection, focusing on CNN architectures (DenseNet, ResNet) and achieving up to 92% AUC on multi-label chest X-ray datasets.
- Investigated fairness and subgroup bias in multi-task and domain-adversarial (DANN) models; used PCA on penultimate layer embeddings to visualize demographic disparities (scikit-learn).
- Fine-tuned foundation models through transfer learning on large-scale radiographic datasets (e.g., CheXpert, MIMIC-CXR), and compared bias in 5+ datasets using Matplotlib, NumPy and Pandas.
- Ran distributed training on HPC clusters using Slurm/PBS job scripting (Linux, Bash).

Inova Health

Washington D.C., USA

Medical Assistant

June 2021 – June 2022

- Facilitated doctor-patient communication, scribed, administered blood tests, and conducted EKGs in a high-volume primary care facility of 3000+ patients. Triaged 30+ patients per day and maintained electronic medical records.

EDUCATION

Imperial College London

October 2022 – October 2023

MSc Computer Science

GPA: Merit; Thesis: Distinction

College of William & Mary

August 2017 – May 2021

BSc Neuroscience

GPA: 3.8/4.0; Dean's Honor List, Summa Cum Laude

- Research Assistant (Supervisor: Dr. Jennifer Bestman); Neural tectal cell analysis (Fiji/ImageJ)
- Research Intern, Rutgers University — PI: Dr. Monica Driscoll (Molecular Biology), May–June 2019
- Research Intern, CECAD Uni Cologne — PI: Dr. Björn Schumacher (Genome Stability in Aging), Jun–Jul 2018

PUBLICATIONS

Hybrid Deep Learning for Traceability and Classification of Industrial Slate Tiles

June 2026

- Soren Antebi, Rafet Sifa et al. IEEE IJCNN 2026. (Accepted)

PROJECTS

Medical Claims Agent | LangGraph, Langfuse, OTel, Claude SDK

August 2026

- Architected LangGraph medical claims agent with deterministic override gates, typed tool schema, and verbatim-excerpt grounding checks; traced with Langfuse/OpenTelemetry.
- Built deterministic eval harness on a stratified golden set, and designed a shadow-mode rollout workflow.

SKILLS

Programming Languages: C/C++, Python, JavaScript, HTML/CSS

Tools & Technologies: Linux/Unix, CMake/Make, Qt, Docker, Git, Slurm, Jupyter, MQTT, Jira, GCP

Frameworks & Libraries: PyTorch/Lightning, Libtorch, HuggingFace Transformers, OpenCV, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, React, Bootstrap, ONNXRuntime, OpenVINO, LLMs/Agents

Languages: Fluent in English and German